

CSE105/209A: Modern Algorithmic Toolbox*, PSET-2

Ian Chi 2253175

1

1.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (1)$$

subtract λ from diagonal

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} \quad (2)$$

the characteristic polynomial is

$$\lambda(\lambda(\lambda^2 - 1) - \lambda) + (-1(\lambda^2 - 1)) = \lambda^4 - 3\lambda^2 + 1 = (\lambda^2 - \lambda - 1)(\lambda^2 + \lambda - 1) \quad (3)$$

which gives roots of

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_2 &= \frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_3 &= -\frac{1}{2} + \frac{\sqrt{5}}{2} \\ \lambda_4 &= \frac{1}{2} + \frac{\sqrt{5}}{2} \end{aligned} \quad (4)$$

*PI: Vaggos Chatziafratis

to find eigenvalues, find the null space after subtracting λI from the diagonal of A .

$$(A - \lambda I)x = 0 = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} x = 0 \quad (5)$$

$$\begin{aligned} -\lambda x_1 + x_2 &= 0 \\ x_1 - \lambda x_2 + x_3 &= 0 \\ x_2 - \lambda x_3 + x_4 &= 0 \\ x_3 - \lambda x_4 &= 0 \end{aligned} \quad (6)$$

if we set x_1 to 1,

$$\begin{aligned} x_2 &= \lambda \\ x_3 &= \lambda^2 - 1 = (\lambda - 1)(\lambda + 1) \\ x_4 &= \lambda - 1/\lambda \end{aligned} \quad (7)$$

note

$$\left[\frac{1}{2}(\pm 1 \pm \sqrt{5}) - 1 \right] \left[\frac{1}{2}(\pm 1 \pm \sqrt{5}) + 1 \right] = \frac{1}{2}(\mp 1 \mp \sqrt{5}) - 1 \quad (8)$$

$$\left[\frac{1}{2}(\pm 1 \mp \sqrt{5}) - 1 \right] \left[\frac{1}{2}(\pm 1 \mp \sqrt{5}) + 1 \right] = \frac{1}{2}(\mp 1 \pm \sqrt{5}) - 1 \quad (9)$$

$$\left[\frac{1}{2}(1 \pm \sqrt{5}) - 1 \right] - \frac{1}{\left[\frac{1}{2}(1 \pm \sqrt{5}) - 1 \right]} = 1 \quad (10)$$

$$\left[\frac{1}{2}(-1 \pm \sqrt{5}) - 1 \right] - \frac{1}{\left[\frac{1}{2}(-1 \pm \sqrt{5}) - 1 \right]} = -1 \quad (11)$$

for λ_1 ,

$$\begin{aligned}
v_1 &= \left(1, -\frac{1}{2} - \frac{\sqrt{5}}{2}, \frac{1}{2} + \frac{\sqrt{5}}{2}, -1\right) \\
v_2 &= \left(1, \frac{1}{2} - \frac{\sqrt{5}}{2}, -\frac{1}{2} + \frac{\sqrt{5}}{2}, 1\right) \\
v_3 &= \left(1, -\frac{1}{2} + \frac{\sqrt{5}}{2}, \frac{1}{2} - \frac{\sqrt{5}}{2}, -1\right) \\
v_3 &= \left(1, \frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{1}{2} - \frac{\sqrt{5}}{2}, 1\right)
\end{aligned} \tag{12}$$

2.

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \quad (13)$$

$$B - \lambda I = \begin{pmatrix} 1-\lambda & -1 & 0 & 0 \\ -1 & 2-\lambda & -1 & 0 \\ 0 & -1 & 2-\lambda & -1 \\ 0 & 0 & -1 & 1-\lambda \end{pmatrix} \quad (14)$$

the characteristic polynomial of B is

$$(1-\lambda)((2-\lambda)((2-\lambda)(1-\lambda)-1)-(1-\lambda))+((-2-\lambda)(1-\lambda))+1) \quad (15)$$

which is equal to $x(x-2)(x^2-4x+2)$ which gives

$$\begin{aligned} \lambda_0 &= 0 \\ \lambda_1 &= 2 - \sqrt{2} \\ \lambda_2 &= 2 \\ \lambda_3 &= 2 + \sqrt{2} \end{aligned} \quad (16)$$

for eigenvectors, we know $v_0 = (1, 1, 1, 1)$

we can also guess $v_2 = (1, -1, -1, 1)$ and it verifies to be correct.

the final two we can also guess, the diagonal is $1 \pm \sqrt{2}, \pm\sqrt{2}, \pm\sqrt{2}, 1 \pm \sqrt{2}$

$$v_1 = (-1, 1 - \sqrt{2}, -1 + \sqrt{2}, 1) \quad (17)$$

$$v_3 = (-1, 1 + \sqrt{2}, -1 - \sqrt{2}, 1) \quad (18)$$

3.

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad (19)$$

one can guess the main eigenvector $(1, 1, 1, 1)$ with eigenvalue 3.

we can also reason that the remaining 3 eigenvalues must be -1. a vector that does not have eigenvalue 3 must be orthogonal to the all ones vector. suppose the eigenvector is orthogonal, its components must sum to 0. the matrix C will take each component i and map it to the sum of the other components which must equal the negative of the component. Thus, any

vector that is orthogonal to $(1, 1, 1, 1)$ must be an eigenvector of with eigenvalue -1.

Here is a valid set of eigenvectors for eigenvalues 1 through 3.

$$\begin{aligned} v_1 &= (1, 0, 0, -1) \\ v_2 &= (0, 1, -1, 0) \\ v_3 &= (1, -1, 0, 0) \end{aligned} \tag{20}$$

note, these were choosen because they span the remaining 3 dimensions and their components sum to 0. They are not necessarily orthogonal, but they can be made orthogonal since spanning vectors can be converted to orthonormal spanning vectors.

4.

$$D = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix} \tag{21}$$

we know the all 1s vector has eigenvalue 0.

we can use a similar reasoning to the last question.

any eigenvector with eigenvalue nonzero (which we know all eigenvectors must be nonzero other than λ_0) must able to be expressed as orthogonal vectors to the all ones vector, i.e. the components must sum to 0. take v to be an arbitrary vector orthogonal to the all ones. $(Cv)_i$ for any component i is equal to $3v_i - \sum_{j \neq i} v_j = 4v_i$. thus any v orthogonal to the all ones is an eigenvector for D .

$$\begin{aligned} v_1 &= (1, 0, 0, -1) \\ v_2 &= (0, 1, -1, 0) \\ v_3 &= (1, -1, 0, 0) \end{aligned} \tag{22}$$

eigenvectors remain the same on complete graphs of n nodes, that being $n - 1$ eigenvectors are eigenvalue n .

5. adjacency

we can solve the two block matrices seperately since the vectors this matrix acts on can be broken into two orthonormal spaces.

for the complete graph, we know

$$\lambda_0 = 0, \lambda_{2,3} = -1$$

$$\begin{aligned}
v_0 &= (1, 1, 1, 0, 0, 0) \\
v_1 &= (1, 0, -1, 0, 0, 0) \\
v_2 &= (-1, 2, -1, 0, 0, 0)
\end{aligned} \tag{23}$$

for the path graph,

$$F_a = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{24}$$

we can guess

$v_3 = (0, 0, 0, 1, 0, -1)$ which has eigenvalue 0.

there is also $v_4 = (0, 0, 0, 1, \sqrt{2}, 1)$ which has eigenvalue $\sqrt{2}$. (if you guess 1,2,1 you see that the output is 2,2,2 so scaling down the center element gives you the eigenvector)

the final eigenvector is $v_5 = (0, 0, 0, 1, -\sqrt{2}, 1)$ since we know how polynomials split, the eigenvalue must include $\lambda_5 - \sqrt{2}$ so from this we find the eigenvector.

laplacian

we know $v_0, v_1 = (1, 1, 1, 0, 0, 0), (0, 0, 0, 1, 1, 1)$ w eigenvalue 0 since they are disconnected.

we also know that there exists two eigenvectors w eigenvalue 3 in the G_1 component

$$v_2, v_3 = (1, 0, -1, 0, 0, 0), (-0.5, 1, -0.5, 0, 0, 0)$$

the final two can be found by finding the eigenvectors for the smaller 3x3 again, we can guess for the matrix

$$F_l = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \tag{25}$$

$$\begin{aligned}
v_4 &= (0, 0, 0, 1, 0, -1) \\
v_5 &= (0, 0, 0, 1, -2, 1)
\end{aligned} \tag{26}$$

$$\lambda_4 = 1, \lambda_5 = 3$$

2

consider the graph consisting of two cliques of size 3 and one of size 2.

assume vertex 1 to 3, 4 to 6, 7,8 are connected. we can write this as a block matrix with the diagonal being 3, 3, 3, 3, 3, 2, 2 and the off diagonals in each block matrix being -1 .

the eigenvectors for this that have eigenvalue 0 would be

$$\begin{aligned} v_0 &= (1, 1, 1, 0, 0, 0, 0, 0) \\ v_1 &= (0, 0, 0, 1, 1, 1, 0, 0) \\ v_2 &= (0, 0, 0, 0, 0, 0, 1, 1) \end{aligned} \tag{27}$$

we know that since the matrix is block diagonal, we chose the vectors carefully so that the product between the vector and the matrix is only affected by a specific block matrix. since each block matrix is like its own separate graph, we know the all 1s vectors on each connected component is an eigenvector with eigenvalue 0.

3

1. instantiate an $n \times n$ matrix A of all zeros.
2. for each node i , choose two distinct nodes j, k that i is connected to. increment the $A_{j,k}$ by 1.
3. if any entry in A is greater than 2, then there exists a cycle.

runtime:

each item above takes n^2 time and are not nested thus the runtime is n^2 .

correctness:

$A_{i,j}$ can only have an entry greater than two iff at least two distinct vertices α, β are neighbors to both i, j . This means there is a 4 cycle on i, α, j, β .

on vertex a , each loop searches for all possible wedges with center point a and increments the entry corresponding to the endpoints of the wedge. In the language of loop invariants, after each loop in item 2, the value of $A_{i,j}$ counts the number of wedges with endpoints i, j that can be formed using only vertices before the index of the loop.

in the end, since every possible center point of wedges was checked, A exactly counts the number of wedges with endpoints i, j . two wedges with distinct center points must mean there is a cycle of length 4.

4

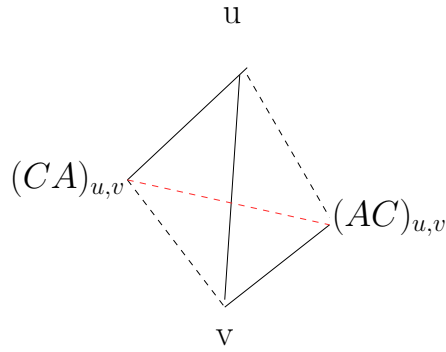
note

$$(CA)_{i,j} = \sum_k C_{i,k} A_{k,j} \quad (28)$$

this value counts how many "paths missing one edge" that end on j there are.

$$(AC)_{i,j} = \sum_k C_{i,k} A_{k,j} \quad (29)$$

this value counts how many "paths missing one edge" that start on j .



assume the red dotted line is not an edge. in this case, this motif (path on 4 vertices) gets counted exactly once with the uv edge.

if the red dotted line is an edge, each edge contributes once to the sum, which is why cycles on 4 nodes get counted 4 times.

i.e. the cycle has 4 paths of length 3 NOT up to isomorphism and since the equation counts paths of length 3, it each cycle shows up 4 times.

5

In [147...

```

import numpy as np

def cycle_matrix(n):
    D = np.diag([2] * n)

    A = np.zeros((n, n), dtype=int)
    for i in range(n):
        A[i, (i + 1) % n] = 1
        A[i, (i - 1) % n] = 1

    L = D - A
    return L

def wheel_matrix(n):
    D = np.diag([3] * n)

    A = np.zeros((n, n), dtype=int)
    for i in range(n-1):
        A[i, (i + 1) % (n-1)] = 1
        A[i, (i - 1) % (n-1)] = 1

    for i in range(n):
        A[i, n-1] = 1
        A[n-1, i] = 1

    L = D - A
    L[n-1, n-1] = 9

    return L

def list_eig(mat):
    eigenvalues, eigenvectors = np.linalg.eig(mat)

    print("Matrix:")
    print(mat)
    print("\nEigenvalues:")
    for i, val in enumerate(eigenvalues):
        print(f"λ{i+1}: {val:.4f}")
    print("\nEigenvecs:")
    for i, val in enumerate(eigenvectors):
        print(f"eigvec{i+1}: {val}")
    return eigenvectors

# if __name__ == "__main__":
#     list_eig(cycle_matrix(10))
#     # list_eig(wheel_matrix(10))

```

cycle

```
In [148... cycle_eig = list_eig(cycle_matrix(10))
```

Matrix:

```
[[ 2 -1  0  0  0  0  0  0  0 -1]
 [-1  2 -1  0  0  0  0  0  0  0]
 [ 0 -1  2 -1  0  0  0  0  0  0]
 [ 0  0 -1  2 -1  0  0  0  0  0]
 [ 0  0  0 -1  2 -1  0  0  0  0]
 [ 0  0  0  0 -1  2 -1  0  0  0]
 [ 0  0  0  0  0 -1  2 -1  0  0]
 [ 0  0  0  0  0  0 -1  2 -1  0]
 [ 0  0  0  0  0  0  0 -1  2 -1]
 [-1  0  0  0  0  0  0  0 -1  2]]
```

Eigenvalues:

```
λ1: -0.0000
λ2: 0.3820
λ3: 1.3820
λ4: 2.6180
λ5: 4.0000
λ6: 3.6180
λ7: 3.6180
λ8: 0.3820
λ9: 1.3820
λ10: 2.6180
```

Eigenvecs:

```
eigvec1: [-0.31622777 -0.4472136 -0.4472136 -0.4472136  0.31622777 -0.4472136
 0.06480388  0.01396596 -0.00566843 -0.00933744]
eigvec2: [-0.31622777 -0.3618034 -0.1381966  0.1381966 -0.31622777  0.3618034
 0.20766368 -0.25143865  0.4235396 -0.42234726]
eigvec3: [-0.31622777 -0.1381966  0.3618034  0.3618034  0.31622777 -0.1381966
-0.40081078 -0.42080224  0.26743029  0.2703624 ]
eigvec4: [-0.31622777  0.1381966  0.3618034 -0.3618034 -0.31622777 -0.1381966
 0.44086178 -0.42943368 -0.25825859  0.2552541 ]
eigvec5: [-0.31622777  0.3618034 -0.1381966 -0.1381966  0.31622777  0.3618034
-0.31251857 -0.27403605 -0.42704288 -0.42811811]
eigvec6: [-0.31622777  0.4472136 -0.4472136  0.4472136 -0.31622777 -0.4472136
 0.06480388 -0.01396596 -0.00566843  0.00933744]
eigvec7: [-0.31622777  0.3618034 -0.1381966 -0.1381966  0.31622777  0.3618034
 0.20766368  0.25143865  0.4235396  0.42234726]
eigvec8: [-0.31622777  0.1381966  0.3618034 -0.3618034 -0.31622777 -0.1381966
-0.40081078  0.42080224  0.26743029 -0.2703624 ]
eigvec9: [-0.31622777 -0.1381966  0.3618034  0.3618034  0.31622777 -0.1381966
 0.44086178  0.42943368 -0.25825859 -0.2552541 ]
eigvec10: [-0.31622777 -0.3618034 -0.1381966  0.1381966 -0.31622777  0.3618034
-0.31251857  0.27403605 -0.42704288  0.42811811]
```

wheel

```
In [149... wheel_eig = list_eig(wheel_matrix(10))
```

Matrix:

```
[[ 3 -1  0  0  0  0  0  0 -1 -1]
 [-1  3 -1  0  0  0  0  0  0 -1]
 [ 0 -1  3 -1  0  0  0  0  0 -1]
 [ 0  0 -1  3 -1  0  0  0  0 -1]
 [ 0  0  0 -1  3 -1  0  0  0 -1]
 [ 0  0  0  0 -1  3 -1  0  0 -1]
 [ 0  0  0  0  0 -1  3 -1  0 -1]
 [ 0  0  0  0  0  0 -1  3 -1 -1]
 [-1  0  0  0  0  0  0 -1  3 -1]
 [-1 -1 -1 -1 -1 -1 -1 -1 -1  9]]
```

Eigenvalues:

λ_1 : 10.0000

λ_2 : 0.0000

λ_3 : 1.4679

λ_4 : 2.6527

λ_5 : 4.8794

λ_6 : 4.0000

λ_7 : 4.8794

λ_8 : 4.0000

λ_9 : 1.4679

λ_{10} : 2.6527

Eigenvecs:

```
eigvec1: [-0.10540926  0.31622777  0.47140452  0.47140452 -0.47140452  0.47140452
 -0.0594396 -0.06678194 -0.00477513 -0.07723751]
eigvec2: [-0.10540926  0.31622777  0.36111681  0.08185854  0.44297535 -0.23570226
 -0.10408807 -0.37073995  0.29933948 -0.47138121]
eigvec3: [-0.10540926  0.31622777  0.08185854 -0.44297535 -0.36111681 -0.23570226
 0.25506119  0.43752188  0.46338982 -0.08647147]
eigvec4: [-0.10540926  0.31622777 -0.23570226 -0.23570226  0.23570226  0.47140452
 -0.37527016 -0.06678194  0.41061491  0.44134998]
eigvec5: [-0.10540926  0.31622777 -0.44297535  0.36111681 -0.08185854 -0.23570226
 0.45021601 -0.37073995  0.16570872  0.23975071]
eigvec6: [-0.10540926  0.31622777 -0.44297535  0.36111681 -0.08185854 -0.23570226
 -0.47085917  0.43752188 -0.15673442 -0.35808543]
eigvec7: [-0.10540926  0.31622777 -0.23570226 -0.23570226  0.23570226  0.47140452
 0.43470976 -0.06678194 -0.40583978 -0.36411247]
eigvec8: [-0.10540926  0.31622777  0.08185854 -0.44297535 -0.36111681 -0.23570226
 -0.34612794 -0.37073995 -0.4650482  0.2316305 ]
eigvec9: [-0.10540926  0.31622777  0.36111681  0.08185854  0.44297535 -0.23570226
 0.21579798  0.43752188 -0.3066554  0.4445569 ]
eigvec10: [ 9.48683298e-01  3.16227766e-01  1.98803348e-16  1.51592507e-17
 1.76144701e-16 -1.29013861e-16  4.51293553e-17  2.29912603e-17
 1.42264122e-17  1.37648139e-17]
```

plots

In [150...

```
import matplotlib.pyplot as plt

def plot_spectral_embedding(v2, v3, num1, num2, title=""):

    norm_v2 = np.linalg.norm(v2)
    norm_v3 = np.linalg.norm(v3)
```

```

# Normalize each vector
v2_normalized = v2 / norm_v2
v3_normalized = v3 / norm_v3

# Use these directly as coordinates
x = v2_normalized
y = v3_normalized

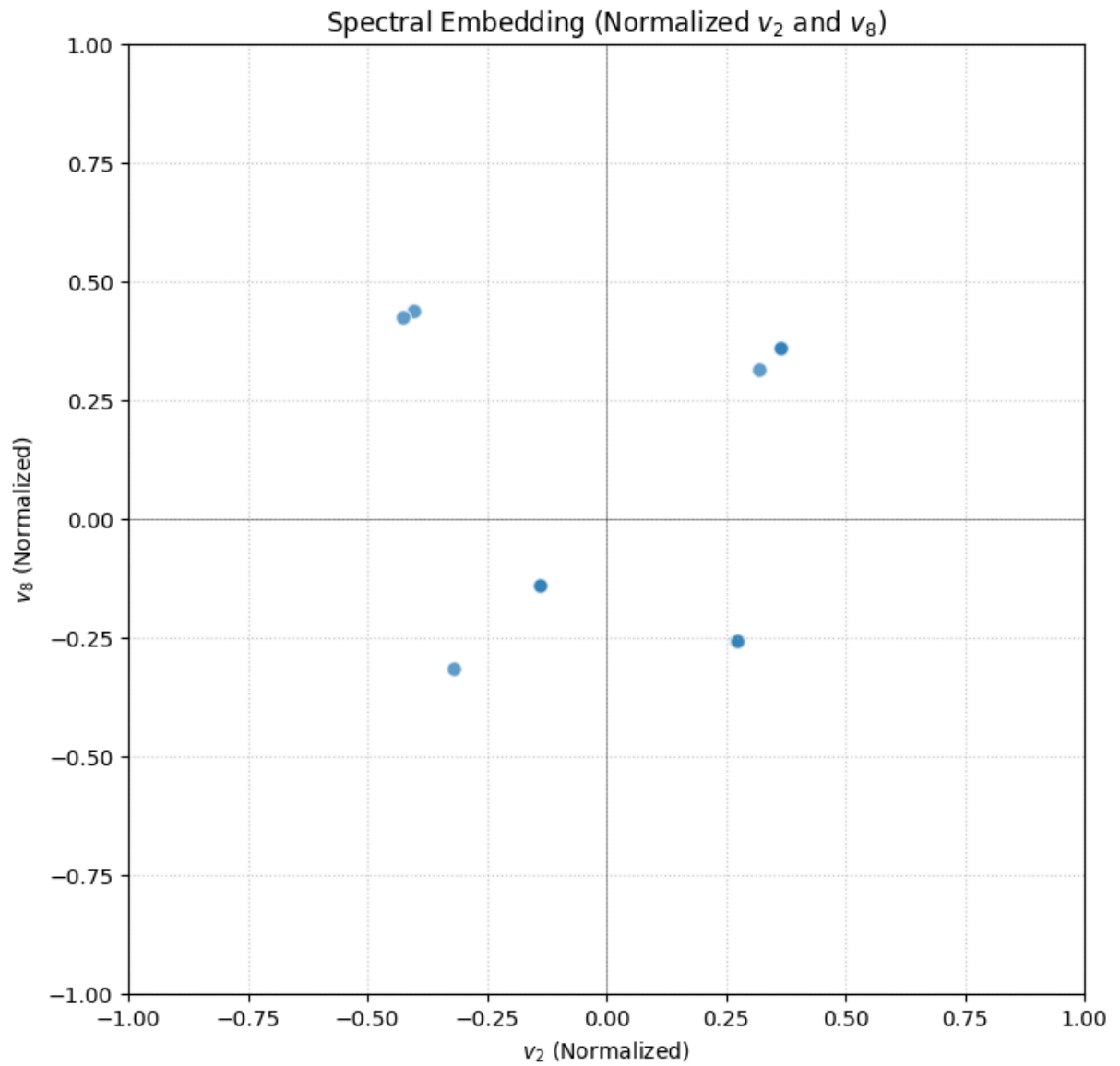
plt.figure(figsize=(8, 8))
plt.xlim(-1, 1)
plt.ylim(-1, 1)
plt.scatter(x, y, alpha=0.7, edgecolors='w', s=50)

plt.axhline(0, color='black', linewidth=0.5, alpha=0.5)
plt.axvline(0, color='black', linewidth=0.5, alpha=0.5)
plt.title(f"Spectral Embedding (Normalized  $v_{\text{num1}}$  and  $v_{\text{num2}}$ )")
plt.xlabel(f" $v_{\text{num1}}$  (Normalized)")
plt.ylabel(f" $v_{\text{num2}}$  (Normalized)")
# plt.axis('equal') # Maintain aspect ratio
plt.grid(True, linestyle=':', alpha=0.6)
plt.show()

```

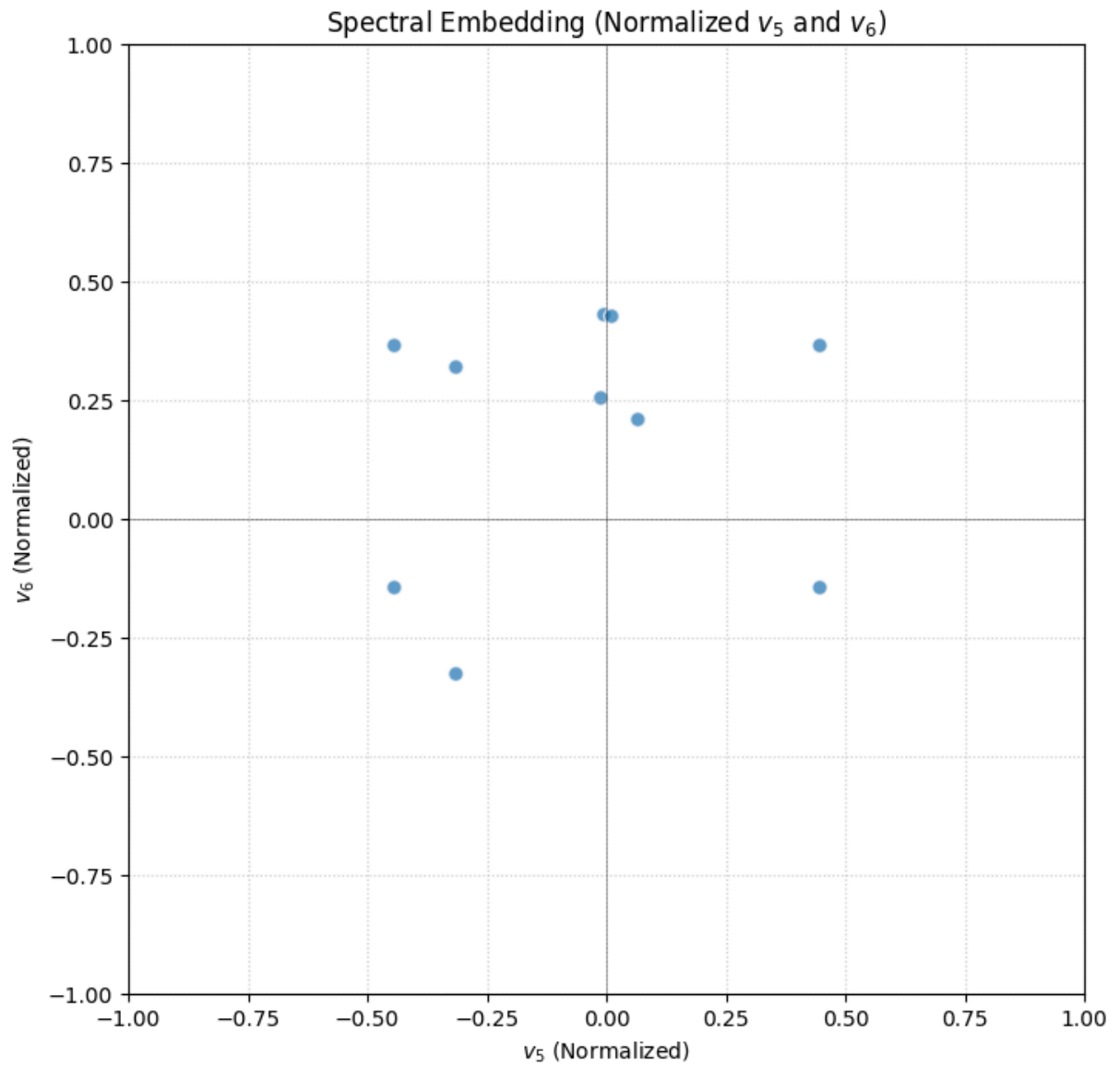
cycle min eigs

In [151... plot_spectral_embedding(cycle_eig[2], cycle_eig[8], 2, 8)



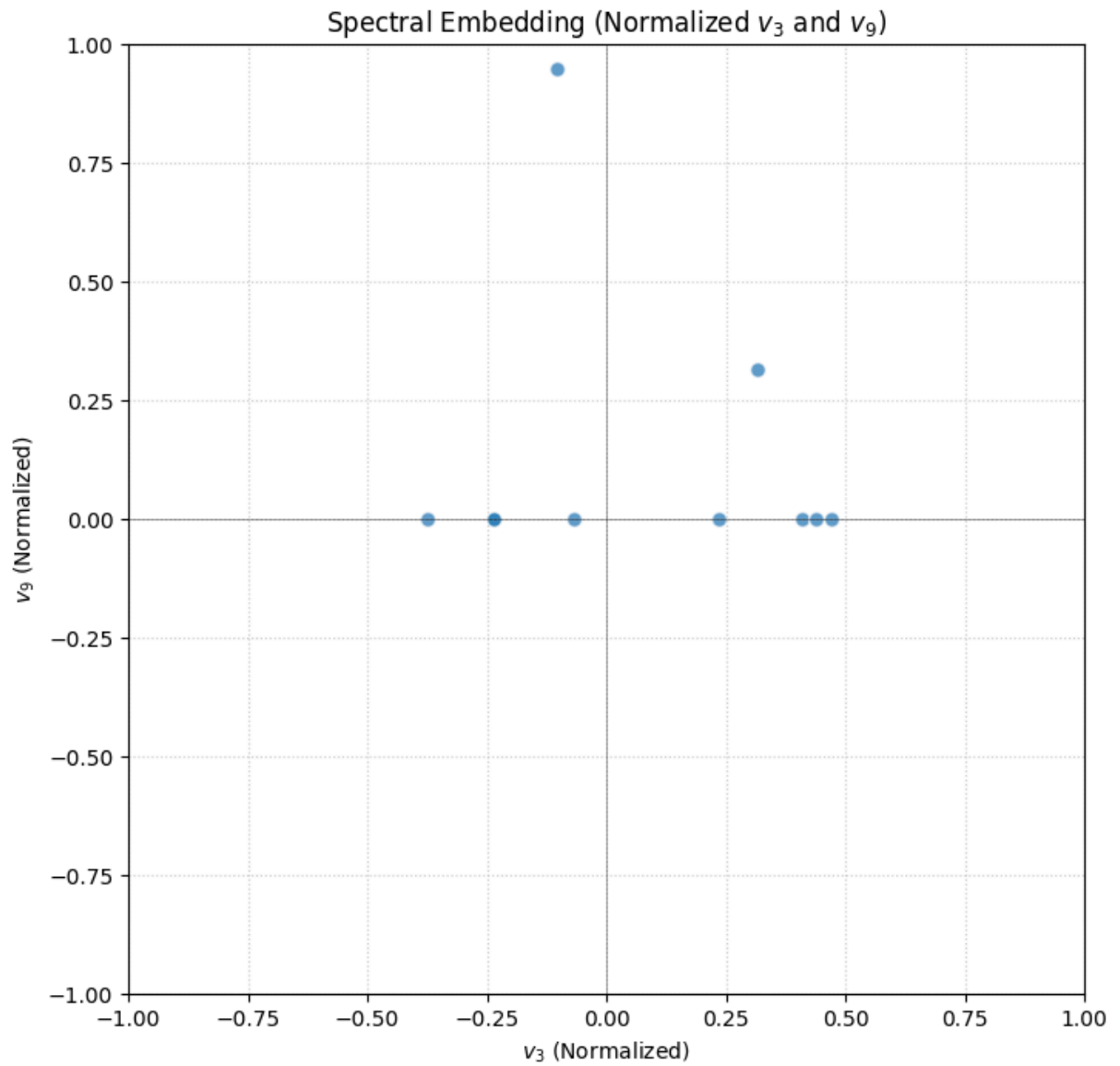
cycle max eigs

```
In [152... plot_spectral_embedding(cycle_eig[5],cycle_eig[6],5,6)
```



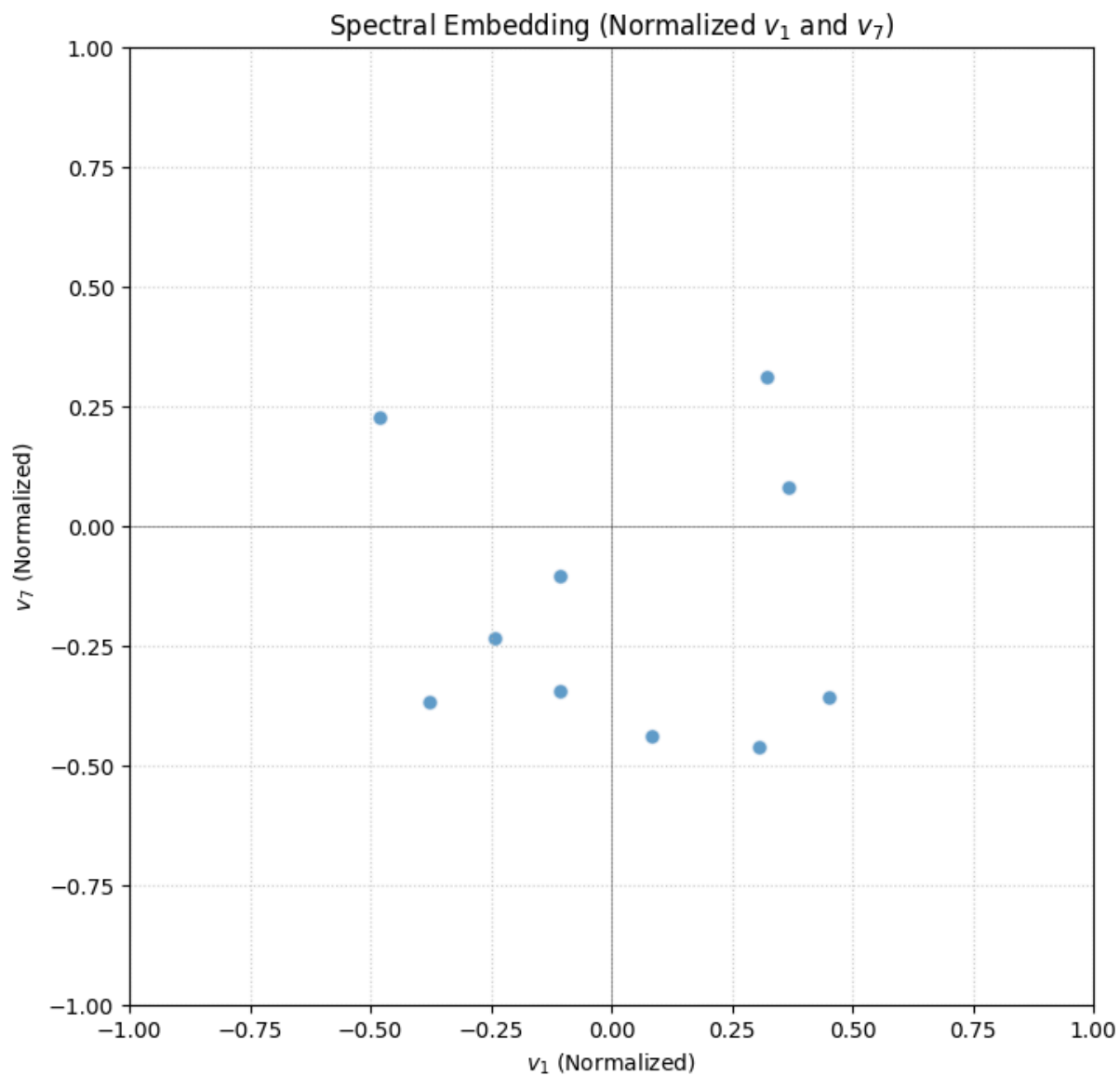
wheel min eigs

```
In [153... plot_spectral_embedding(wheel_eig[3],wheel_eig[9],3,9)
```



wheel max eigs

```
In [154... plot_spectral_embedding(wheel_eig[1],wheel_eig[7],1,7)
```



the spectral embeddings with small eigenvectors are allowing one to see how the graph is connected. info not captured by second eig is captured partially by v_3 .

for large eigenvectors, the eigenvectors show how separated two vertices are.